MILITARY SPECIFICATION

RECEIVING SET, RADIO
AN/ARW-79

This specification has been approved
by the Bureau of Naval Weapons,
Department of the Navy

1.0 SCOPE

1.1 Scope - The equipment covered by this specification shall be airborne equipment for radio control of drones and target aircraft.

1.2 Classification - The Radio Receiving Set AN/ARW-79 shall be of one type and shall consist of the following.

Unit	Type	Max. Allowable Size (Inches)	Max. Allowable Weight (lbs.)	Appl. Para.
Receiver, Radio	R-1182/ARW-79	4.25x6.0x2.0	1.5	3.4.1.2, 3.4.1.3
Decoder, Command Signals	KY-488/ARW-79	5.62x7.25x5.5	4.25	3.4.2.2, 3.4.2.3
Decoder Command Signals	KY-489/ARW-79	3.2x5.2x2.0	1.5	3.4.3.2, 3.4.3.3
Antenna	AS-1391/ARW-79	7.25 long	.20	3.4.4.2, 3.4.4.3

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1.3 Associated Equipment - The Radio Receiving Set shall operate with the following associated equipments or equivalents which are not to be supplied as part of this equipment:

Radio Transmitting Set	AN/URW-14
Coder, Audio	KY-133/ARW-66
Coder, Audio	KY-294/USD-1 (Mod.)
Control, Radio	C-1395/ARW-66
Control, Radio	C-1396/ARW-66
Control, Radio	C-2939/USD-1 (Mod.)
Servo Actuator	D-9
Gyro Vertical	K-7

2.0 APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-N-25076	Nameplate, Identification, Metallic Foil, Adhesive-backed
MIL-D-70327	Drawings; Engineering and Associated Lists

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STANDARDS

MIL-STD-129	Marking for Shipment
MIL-STD-449A	Radio Frequency Spectrum Characteristics
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization
WR-29	Naval Weapon Requirements, Compatibility Frequency Allocation and Equipment Spectrum Signature

2.2 Availability of Documents - When requesting specifications, standards, drawings and publication, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

2.3 Precedence - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

2.3.1 Contract - The contract shall have precedence over any specification.

2.3.2 This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviations from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

2.3.3 Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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3.0 REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and materials requirements shall conform to Specification MIL-E-5400.

(b) Non-repairable sub-assemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the sub-assembly and the amount of circuitry included therein shall be approved by the procuring activity. Non-repairable sub-assemblies must be reliable.

(c) When previously produced models of this equipment did not use non-repairable sub-assemblies, the design shall not be changed to employ non-repairable sub-assemblies without the approval of the procuring activity.

3.1.1 Non-Standard Parts and Materials Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.1.3 Relays - Relay contacts shall be rated at not less than 2.0 amperes non-inductive at 28 vdc.

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3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.1 Total Weight - The weight of the equipment shall be a minimum consistent with good design and shall not exceed 7.45 pounds.

3.2.2 Cables and Connections - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.3 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.4 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.5 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal Ground (25" to 31" Hg)
Vibration	None
Humidity	Room ambient up to 90% relative humidity

3.2.6 Service Conditions - The equipment shall meet the requirements of this specification under any of the environmental service conditions or natural combination of these conditions as specified in MIL-T-5422 for Class 1 equipment with the following exceptions.

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3.2.6.1 Immersion - The equipment shall withstand immersion in water to a depth of three feet for a period of two hours without leakage.

3.2.7 Nameplates and Nomenclature - Nameplate format and nomenclature for equipment identification shall be in accordance with Specification MIL-N-18307. Nameplate material shall be in accordance with Specification MIL-N-25076.

3.2.8 Electrical Receptacles - The equipment shall contain the following external receptacles with color coding and identification as indicated:

(a) Antenna, Gremar 5597-1 J101

(b) Gyro, AN 3102C-16S-1S, Blue, J102

(c) Elevator, AN3102C-16S-1S, Yellow, J103

(d) Power, AN3102C-16S-1P, J104

(e) Aileron, AN3102C-16S-1S, Red, J105

(f) Servo Centering, MS3102C-12S-3S, J106

(g) Recovery (parachute), MS3102C-10SL-4P, J107

(h) Accessory Channels, Cannon DBH-25P J109

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Operating Life

3.3.2.1 Operational Stability - The equipment shall operate for 50 hours continuously or intermittently, without necessity for

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readjustment. and without failure.

3.3.2.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 150 hours without requiring removal for bench servicing. Parts requiring servicing or replacement at the end of this interval, in order to renew this service life, shall be specified by the manufacturer.

3.3.2.3 Total Operating Life - The equipment shall have a minimum total operating life of 500 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.3.3 Voltage Limits - The Radio Receiving Set, AN/ARW-79, shall operate from a d-c source whose terminal voltage is within the limits of 22.5 to 30 volts. The ripple voltage shall not exceed 1.5 volts RMS nor transients greater than 70 volts peak-to-peak at a duration of 10 milliseconds, maximum, in accordance with MIL-STD-704. The Radio Receiving Set shall not be damaged by reversal of input voltage polarity.

3.3.4 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed one minute under standard conditions and five minutes at extreme service conditions.

3.3.5 Under-Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returned to within limits.

3.3.6 Overload Protection - All parts and circuits in the equipment which are likely to carry an overload due to poor adjust-

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ment shall be proportional to withstand such overload without failure of the equipment. Fuses or circuit breakers shall not be used. A failure is anything that causes malfunction of the equipment.

3.3.7 Input Current Drain - The Radio Receiving Set, AN/ARW-79, shall operate from a nominal 27.5 vdc source, and the total current drain, excluding associated equipment, shall not be in excess of 1.0 amperes at 27.5 vdc.

3.3.8 Interference Control - The generation of radio interference and conducted noise interference by the equipment shall be controlled so as not to exceed the limits of Specification MIL-I-6181.

3.3.9 Frequency Characteristics

3.3.9.1 Radio Frequency Characteristics - The equipment shall be capable of receiving within the range of 406 to 550 mc at any discrete carrier frequency ±0.01 per cent with a total combined tone modulation of ±300 kc. (±120 kc for each of the two ON-OFF Modulations and ±60 kc for the auxiliary command).

3.3.9.2 Audio Frequency Characteristics - The audio command frequencies are preselected and generated by a KY-133/ARW-66 coder or equivalent. The frequency tolerance of the coding system is ±1%.

3.3.10 Sensitivity - The equipment shall provide reliable reception of distance signals, providing an input amplitude of 10 microvolts is available, for all performance conditions specified herein.

3.3.11 Selectivity - The operation of the equipment shall not be interfered with by RF signals of 10,000 microvolts intensity which are 2 mc or greater above or below the center frequency when a center frequency signal of 20 microvolts intensity is being received.

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3.3.12 Frequency Stability - The r-f and a-f circuitry stability of the equipment shall be such that it operates reliably under all performance conditions specified herein.

3.3.13 Images and Spurious Responses - All images, spurious and parasitic frequencies shall be held to a minimum with an attenuation of at least 60 db below the desired frequency.

3.4 Detail Design Requirements

3.4.1 Radio Receiver R-1182/ARW-79 - The Radio Receiver shall meet the following requirements.

3.4.1.1 Function - The Radio Receiver shall be designed to receive a frequency modulated carrier which is modulated continuously by two proportional command channels and by any one of seven other tones for an ON-OFF type of command.

3.4.1.2 Form Factor and Mounting - The maximum size shall not exceed 4.25 inches wide by 6.0 inches long by 2.0 inches deep including flanges, brackets and connectors. The receiver shall mate with Decoder, Command Signals KY-488/ARW-79 without shock mounts. The receiver shall be waterproof when mated to Decoder, Command Signals KY-488/ARW-79.

3.4.1.3 Weight with Mounting - The total weight of the receiver with mounting provisions shall not exceed 1.5 pounds.

3.4.1.4 Contents of Receiver - The Radio Receiver shall contain all circuitry necessary to receive the RF signals, demodulate and apply the necessary signal to the decoder.

3.4.1.4.1 Crystal Controlled Local Oscillator - A crystal controlled local oscillator shall be provided which shall be frequency stabilized to within $\pm$.01% over the full range of service conditions.

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3.4.1.4.2 IF Amplifiers and Limiters - The IF amplifiers and limiters shall have bandpass characteristics capable of satisfying the sensitivity requirements under paragraphs 3.3.9 and 3.3.10. The design shall be such that optimum performance will be obtained for total carrier deviation of ±300 kc.

3.4.1.5 Power Input - The Radio Receiver shall operate from a nominal 27.5 and/or 17.0 volts dc source, and the current drain shall not be in excess of .015 amperes at 27.5 vdc and .035 at 17.0 volts.

3.4.1.6 Voltage Limits - The Radio Receiver shall operate reliably under all performance conditions specified herein with the following d-c input voltages.

21.0 to 30.0 volts	27.5 nominal
16.0 to 18.0 volts	17.0 nominal

3.4.1.7 Sensitivity - When supplied with an input signal at receiver nominal frequency, frequency modulated ±300 kc at an audio frequency between 7.0 and 20.0 kc, the Radio Receiver shall not require more than 10 microvolts RF input to deliver 6.5 volts peak-to-peak of filtered audio output. The audio output shall be measured free of noise after a narrow band filter with a 1:1 transfer ratio. The filter shall be such that it effects negligible loading on the receiver.

3.4.1.8 Output Impedance - When operated at room temperature with a 100 microvolt input signal with a frequency deviation of ±60 kc at an audio frequency between 7 and 13 kc, the audio output of the Radio Receiver shall not drop to less than 35 per cent when a 50 ohm audio load through a 10 microfarad capacitor is connected directly at pin D of the power connector.

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3.4.1.9 Noise Output - With no RF input signal, the noise output of the receiver shall not exceed 3.25 VRMS when measured with a multimeter model 630-NA (triplet or equivalent) in the output position.

3.4.1.10 Audio Output - When operated with a 100 microvolt input signal with a frequency deviation of ±300 kc at an audio frequency between 7 and 13 kc, the audio output of the Radio Receiver, without load, shall not be less than 13.0 volts peak-to-peak.

3.4.1.11 Antenna Impedance - The Radio Receiver shall be designed to work with antennas of 51 ohms nominal impedance.

3.4.1.12 Electrical Receptacles - The Receiver shall contain a Winchester M4P or equivalent for interconnection to Decoder, Command Signals KY-488/ARW-79.

3.4.2 Command Signals Decoder KY-488/ARW-79 - The command signals decoder shall meet the following requirements.

3.4.2.1 Function - The decoder, when commanded by coded audio signals or by signals from a vertical gyro, or both, shall operate two external servos for control of a pilotless aircraft, shall furnish a carrier loss signal and one ON-OFF signal for parachute release and accessory control respectively.

3.4.2.2 Form Factor and Mounting - The maximum size shall not exceed 5.62 inches wide by 7.25 inches long by 5.5 inches deep excluding mounting provisions and external connectors.

3.4.2.3 Weight with Mounting - The total weight of the decoder with mounting provisions shall not exceed 4.25 pounds.

3.4.2.4 Contents of Command Signals Decoder - The decoder shall contain all circuitry necessary to isolate and apply the audio

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signals to an integral amplifier. The amplifier shall contain all necessary circuitry to permit operation of external equipment. The following circuitry shall be provided.

(a) Audio Filters

(b) Pitch Command Channel and Stabilizing Circuitry

(c) Roll Command Channel and Stabilizing Circuitry

(d) Recovery Channel

(e) Accessory Channel

(f) Regulated Power Supply

(g) Provisions to Accept a Radio Receiver

(h) Provisions to Accept an Accessory Channel Decoder.

3.4.2.5 Commands - The Decoder shall respond to two pulse modulated audio tones, one for proportional pitch attitude control and one for roll attitude control, and one ON-OFF audio tone for accessory control.

3.4.2.5.1 Command Frequencies - The audio frequencies of the command channels shall be as follows:

Accessory Command Channel	7,500 cps
Pitch Command Channel	10,760 cps
Roll Command Channel	13,700 cps

3.4.2.5.2 Command Characteristics - The 10,760 cps and 13,700 cps audio commands shall be pulsed at a 15 cps $\pm$ 5% rate. The pulse length shall be continuously variable from approximately 20 to 80 per cent.

3.4.2.5.3 Command Input - The Decoder shall be designed to operate when supplied with an audio signal of not more than 5.2 volts peak-to-peak for the two proportional command channels and

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not more than 1.6 volts peak-to-peak for the spare command channel from a source having an internal impedance of 50 ohms.

3.4.2.5.4 Pitch Command and Stabilizing Circuitry - The Decoder pitch command and stabilizing circuits shall be so designed that when the specified servos and vertical gyro, or equivalent test set, are connected to the external connectors, the extreme pulse width variation shall command pitch angles of gyro attitude or test set equivalent in conformance with the table below, and intermediate angles, including trim commands, in direct proportion to the pulse duty ratio.

50% duty ratio = $0^{\circ} \pm 3\text{-}1/2^{\circ}$

35% duty ratio = 30° -5° $+8^{\circ}$ Up

65% duty ratio = 30° -5° $+8^{\circ}$ Down

3.4.2.5.5 Roll Command and Stabilizing Circuitry - The Decoder roll command and stabilizing circuits shall be so designed that when the specified servos and vertical gyro, or equivalent test set, are connected to the external connectors, the extreme pulse width variation shall command pitch angles of gyro attitude or test set equivalent in conformance with the table below, and intermediate angles, including trim commands, in direct proportion to the pulse duty ratio.

50% duty ratio = $0^{\circ} \pm 5^{\circ}$

30% duty ratio = 50° -5° $+10^{\circ}$ Left

70% duty ratio = 50° -5° $+10^{\circ}$ Right

3.4.2.5.6 Recovery Command - A "fail-safe" relay shall be provided and its output circuit shall be so arranged that in the event the radio receiver becomes inoperative or the r-f carrier signal is removed from the receiver, or the pitch command tone is

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removed, parachute command shall be initiated after a delay of 0.5 to 1.0 seconds.

3.4.2.5.7 Spare Command - A spare channel circuit shall be provided to operate from a 7,500 cps tone for supplying an accessory command "ground" signal.

3.4.2.6 Servo and Gyro Follow-Up Ratio

3.4.2.6.1 Pitch - The equipment shall be so designed that signals from the gyro or test set equivalent shall position the pitch servo or test set equivalent in conformance with the table below, and intermediate angles, in direct proportion.

0° pitch gyro output	=	center $\pm 5^\circ$ servo deflection
30° up gyro output	=	$18^\circ \pm 5^\circ$ up servo deflection
30° down gyro output	=	$18^\circ \pm 5^\circ$ down servo deflection

3.4.2.6.2 Roll - The equipment shall be so designed that signals from the gyro or test set equivalent shall position the roll servo or test set equivalent in conformance with the table below, and intermediate angles, in direct proportion.

0° roll gyro output	=	center $\pm 5^\circ$ roll servo deflection
40° left gyro output	=	$20^\circ \pm 5^\circ$ left servo deflection
40° right gyro output	=	$20^\circ \pm 5^\circ$ right servo deflection

3.4.2.7 Command Channel Interaction - Interaction between the roll, pitch and spare channel circuits shall not exceed two degrees of servo or test set equivalent deflection.

3.4.2.8 Gyro Information - The decoder shall accept and respond to attitude information from a suitable vertical gyro such as a Type K-7 in both pitch and roll axes.

3.4.2.9 Servo Position Information - The decoder shall utilize the follow-up potentiometer of a servo type D-9 or equivalent.

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3.4.2.10 Electrical Receptacles - The Decoder shall contain the following connectors for interconnection of the Radio Receiving Set AN/ARW-79.

a. Radio Receiver, Winchester M4S P-1

b. Accessory Decoder, Winchester M4P J-108

3.4.2.11 Accessory Power - The Decoder shall be designed to provide power for interconnection of the Radio Receiving Set AN/ARW-79. With a 27.5 volt dc input, the Decoder shall provide no less than 25.5 volts at a load of 15 ma. With a 22.5 to 30.0 volt dc input, the Decoder shall provide 17 $\pm$ 1.0 volts dc at a load of 35 ma.

3.4.3 Command Signals Decoder - KY-489/ARW-79 - The command signals decoder shall meet the following requirements.

3.4.3.1 Function - The decoder, when commanded by audio signals, shall provide six ON-OFF channels for control of accessory equipment.

3.4.3.2 Form Factor and Mounting - The maximum size shall not exceed 3.2 inches wide by 5.2 inches long by 2.0 inches deep, excluding external connectors. The Decoder, Command Signals KY-489 shall mate with Decoder, Command Signals KY-488 without shock mounts. The decoder shall be waterproof when mated to Decoder, Command Signals KY-488/ARW-79.

3.4.3.3 Weight with Mounting - The total weight of the decoder with mounting provisions shall not exceed 1.5 pounds.

3.4.3.4 Commands

3.4.3.4.1 Command Frequencies - The audio frequencies of the accessory command channels shall be as follows:

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Accessory Spare 1	8,460 cps
Accessory Spare 2	9,540 cps
Accessory Spare 3	12,140 cps
Accessory Spare 4	15,450 cps
Accessory Spare 5	17,430 cps
Accessory Spare 6	19,660 cps

3.4.3.4.2 Command Input - The Decoder shall be designed to operate, when supplied with an audio signal at nominal channel frequency of not more than 1.6 volts peak-to-peak from a source having a 50 ohm internal impedance, the desired accessory command channel.

3.4.3.4.3 Command Output - The Decoder shall provide SPDT relay contact closure upon reception of an accessory command audio tone at nominal channel frequency.

3.4.3.5 Power Input - The Decoder shall operate from a nominal 27.5 volt dc source, and the current drain shall not be in excess of .08 amperes with one channel energized.

3.4.3.6 Voltage Limits - The Decoder shall operate reliably under all performance conditions specified herein with an input voltage of 22.5 to 30 volts.

3.4.3.7 Electrical Receptacle - The Decoder shall contain a Winchester M4S or equivalent for interconnection to Decoder, Command Signals KY-488/ARW-79.

3.4.4 Antenna, AS-1391/ARW-79 - The antenna shall meet the following requirements.

3.4.4.1 General - The antenna shall operate with the Receiver, Radio R-1182/ARW-79.

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3.4.4.2 Form Factor and Mounting - The maximum size shall not exceed 7.25 inches long, including mounting provisions. The antenna shall be of the stub type with a BNC connector on the mounting end.

3.4.4.3 Weight with Mounting - The total weight of the antenna with mounting provision shall not exceed .20 pounds.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests

(b) Production Sample Tests

(c) Acceptance Tests

(d) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(a) Contractor's Demonstration Tests

(b) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

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4.2.1.1 Data Submittal - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature ratings and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Fabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipment Submitted - Equipments supplied under the contract shall in all respects including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by this government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first ten production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that

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the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptable tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material

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during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product

(b) Operational Tests

(c) Manufacturing Reliability Tests

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of six (6) hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during

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the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purposes of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the 6-hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failure occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests.

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Equipments shall be selected for sampling tests by Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(a) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(c) Manufacturing reliability test specified in paragraph 4.4.1.3, except that the test duration shall be 150 hours with no failures.

Each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - The special tests shall be conducted on a quantity of equipments for the purpose of checking the effect

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of any design of material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows, except that when a Production Sample is requested and delivered to a government laboratory for test (see paragraph 4.3), the requirement for a special test on an early production equipment may be omitted.

(a) On an early production equipment

(b) On an early equipment after an Engineering change

(c) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(a) Temperature - Altitude Tests

(b) Vibration Test (60 minutes cycling in each of 3 planes)

(c) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

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(a) Determine the cause of failure

(b) Determine if the failure is an isolated case or design defect

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 150 hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following, except that when a production sample equipment is requested and submitted to the procuring activity for test, then the first equipment specified below need not be selected or subjected to this test (equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (-1000-5000 ft.)
Humidity	Room ambient

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - Basic parameters shall be maintained continuously. At approximately 24-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures per MIL-STD-449 and WR-29 data. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including the tube failures. This record shall indicate the following:

(a) Part type number

(b) The circuit reference symbol number

(c) The part function

(d) Name of the manufacturer

(e) Nature of the failure

(f) The number of hours which the part operated prior to failure.

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4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(a) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No items, part or complete

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equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

4.9 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

5.0 PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order. Marking shall be in accordance with MIL-P-17555 and preservation shall be method II specification MIL-P-116 for the receiver and both decoders; and method I C specification MIL-P-116 for the antenna.

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